

Cody Melcher

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Statistics and Data Science GIDP, University of Arizona, Tucson, AZ 85721

EDUCATION

Ph.D. in Statistics & Data Science (Minor: Systems and Industrial Engineering) May 2026 (expected)

University of Arizona, Tucson, AZ

M.S. in Economics

December 2019

University of Arizona, Tucson, AZ

B.A. in Economics

May 2014

Reed College, Portland, OR

RESEARCH INTERESTS

- Min-max (saddle point) optimization
 - First-order and primal-dual algorithms
 - Optimization on embedded and resource-limited devices (TinyML)
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AWARDS AND HONORS

- **Carter Travel Award**, University of Arizona — Oct 2024 (\$600 research travel support)
 - **Honorable Mention for Mathematics Graduate TA**, University of Arizona — Spring 2023
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PUBLICATIONS

Accepted / Under Review

- C. Melcher, Z. Alizadeh, L. Hiett, A. Jalilzadeh, and E. Yazdandoost Hamedani.
“*Semi-infinite Nonconvex Constrained Min-Max Optimization.*”
Accepted in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. (Acceptance rate 24.5%)
arXiv preprint arXiv:2510.12007, 2025.
- C. Melcher, A. Jalilzadeh, and E. Yazdandoost Hamedani.
“*Linear Convergence of a Unified Primal-Dual Algorithm for Convex-Concave Saddle Point Problems with Quadratic Growth.*”
First revision in *Optimization Letters*.
arXiv preprint arXiv:2510.11990, 2025.

Works in Progress

- C. Melcher, A. Jalilzadeh, and E. Yazdandoost Hamedani.
“*Convergence of Primal-Dual Algorithms for Min-Max Semi-Infinite Programs under Concavity.*”
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RESEARCH PROJECTS

Linear Convergence without Strong Monotonicity (Optimization Letters Project)

Developed the first unified primal-dual framework ensuring linear convergence of first-order methods for saddle point problems without assuming strong convexity or strong monotonicity. Introduced the two-sided Quadratic Functional Growth (2-QFG) and two-sided Quadratic Gradient Growth (2-QGG) curvature conditions.

Semi-Infinite Nonconvex Constrained Min-Max Optimization (NeurIPS Project)

Proposed an inexact dynamic barrier primal-dual (iDB-PD) algorithm for semi-infinite constrained problems that enforces feasibility without penalty terms and achieves convergence to an ε -KKT point under Łojasiewicz regularity assumptions.

On-Device Optimization and TinyML Deployment

Implemented and trained deep learning models for MNIST and Multi-MNIST classification, then deployed quantized

models to the Arduino Nano 33 BLE Sense. Extending iDB-PD for real-time constrained learning on embedded hardware under strict memory and precision limits.

PRESENTATIONS

- **NeurIPS (Conference on Neural Information Processing Systems)** Dec 2025
San Diego, CA — Poster presentation of *Semi-infinite Nonconvex Constrained Min-Max Optimization*.
Session: *Optimization and Learning under Uncertainty*
 - **INFORMS Annual Meeting** Oct 2025
Atlanta, GA — Talk: *Semi-infinite Nonconvex Constrained Min-Max Optimization*.
Session: *Structured and Hierarchical Optimization*.
 - **INFORMS Annual Meeting** Oct 2024
Seattle, WA — Talk: *Incorporating Quadratic Functional Growth in Saddle Point Problems*.
Session: *Game Theory and Optimization: Minimax and Variational Inequality Approaches*.
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TEACHING EXPERIENCE Graduate Teaching Assistant, University of Arizona

- **MATH 263: Introduction to Statistics and Biostatistics** Jan 2025 – May 2025
Led recitations, office hours, and assisted in assessment and course design.
 - **MATH 112: College Algebra** Aug 2022 – May 2023
Created and graded homework, led review sessions, and provided curriculum support.
Honorable Mention for Mathematics Graduate TA (Spring 2023).
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PROFESSIONAL ACTIVITIES

- Professional Memberships: INFORMS (Member, 2024–present)
 - Outreach and Mentoring: Presenter, “Optimization, Neural Networks, and TinyML” Summer Workshop for High School Students, University of Arizona (Summer 2025)
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LANGUAGES AND SKILLS

- **Programming:** Python (PyTorch, TensorFlow), R, MATLAB, LaTeX, Stata
- **Technical:** Constrained optimization, saddle-point theory, primal–dual algorithms, embedded ML.